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 Neo Technology



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# Databases are boring!

Amanda @pandamonical Laucher





What does a DB do for  
YOU

















It's a-me,  
Mario!





Algorithms are...



Algorithms are  
AWESOME!

How writing  
applications works



Step 1:  
Write some awesome  
algorithms

Step 2:

I put mock data into  
a data structure that  
is interesting to me



Step 3:

Draw up some fancy  
UI so end users are  
jolly and they keep  
paying me

Step 4:

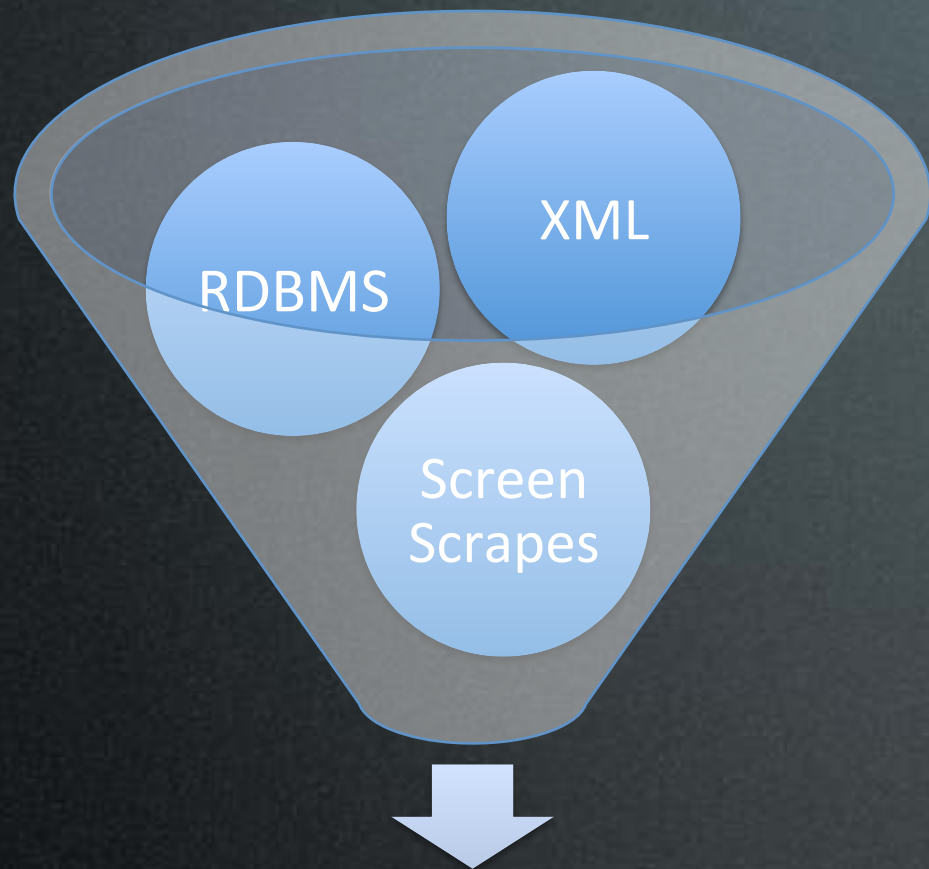
Let some poor  
schmuck connect to  
the DBs and get the  
data in the format I  
need



Step 5:  
Profit!

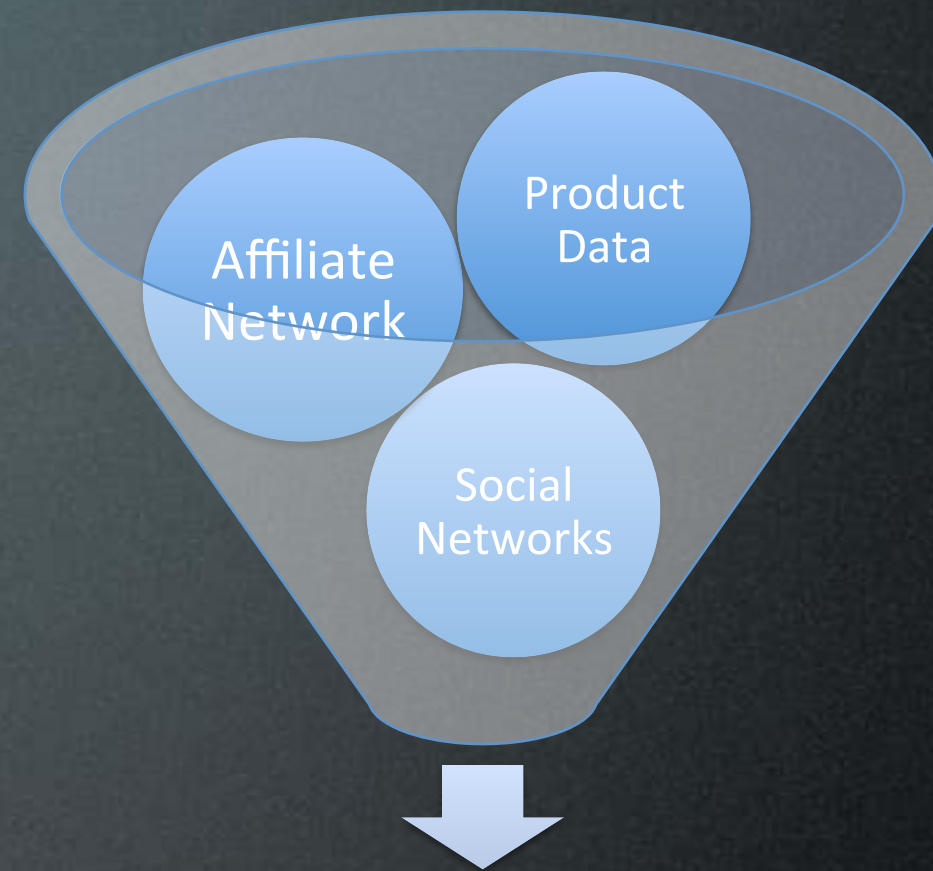
Data persistence has  
always been an  
afterthought





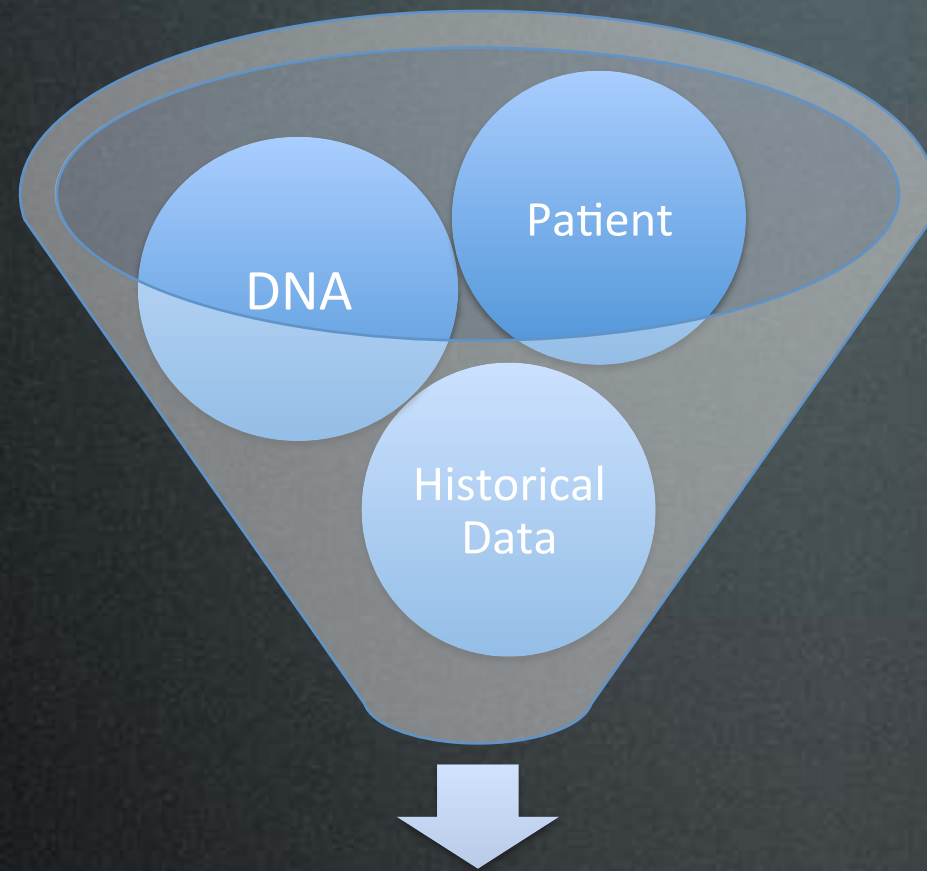
\$\$\$ **BIG DATA** \$\$\$





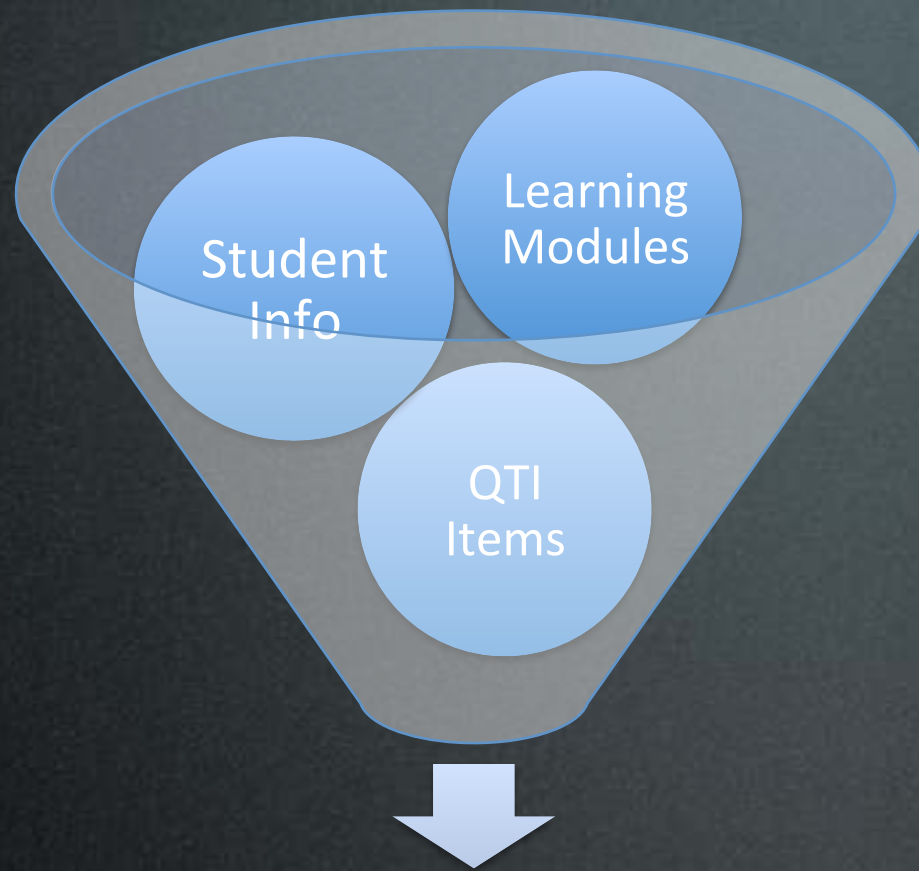
\$\$ Personalized Deals \$\$





\$\$ Cure Cancer \$\$

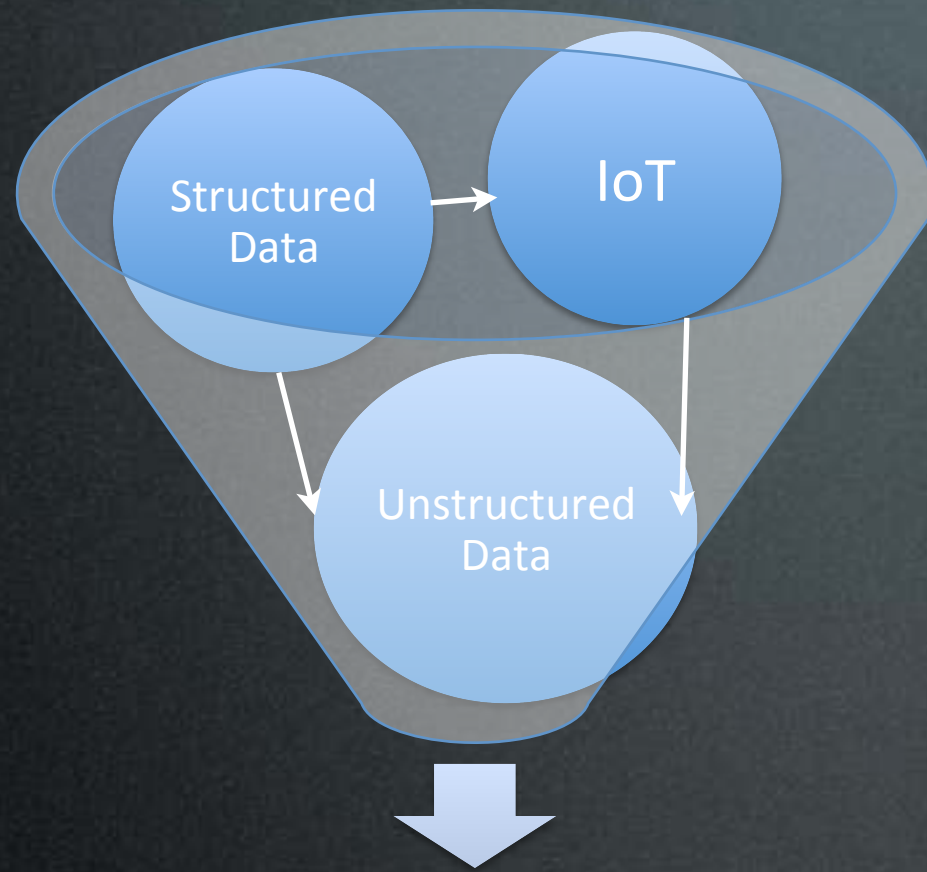




\$\$ Personalized Learning \$\$

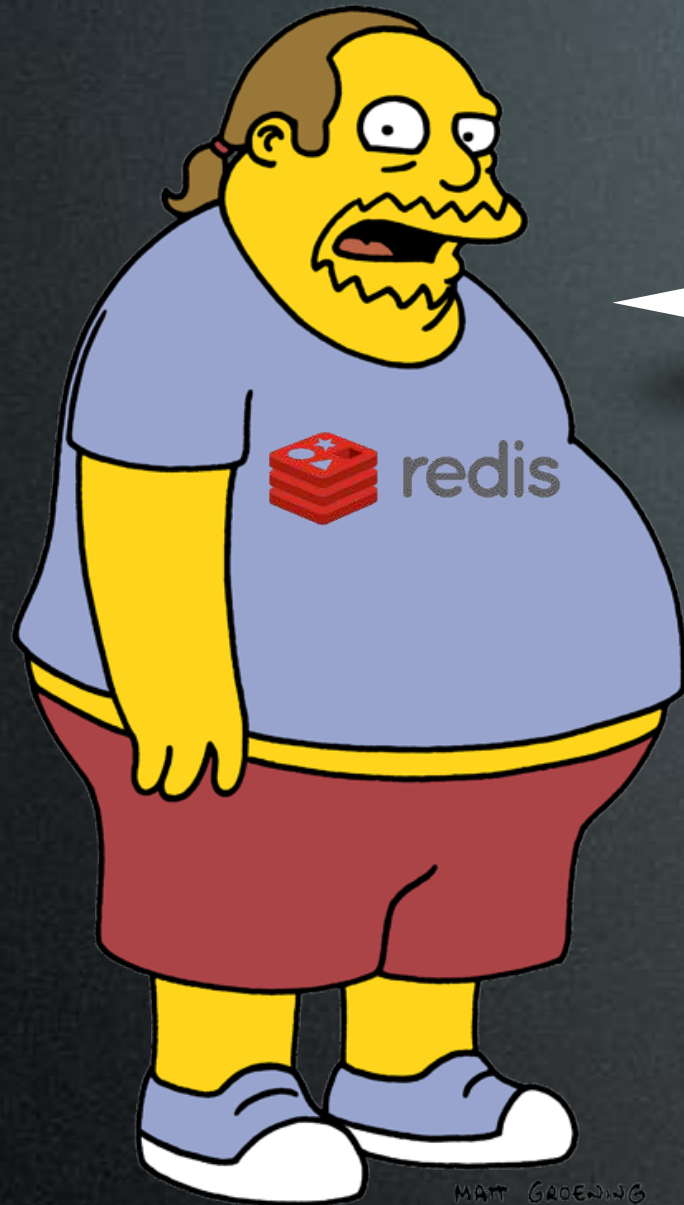






\$\$ Connected Data \$\$

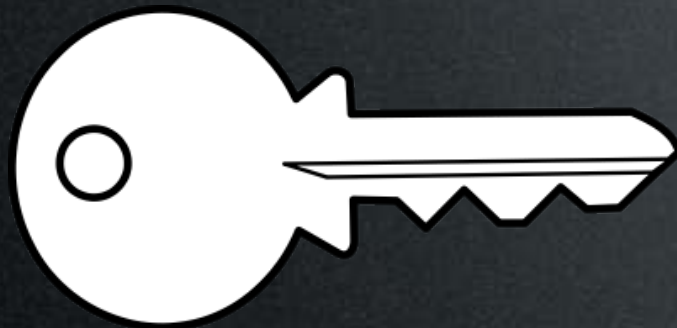




Just use a key value store.  
Best data storage ever!

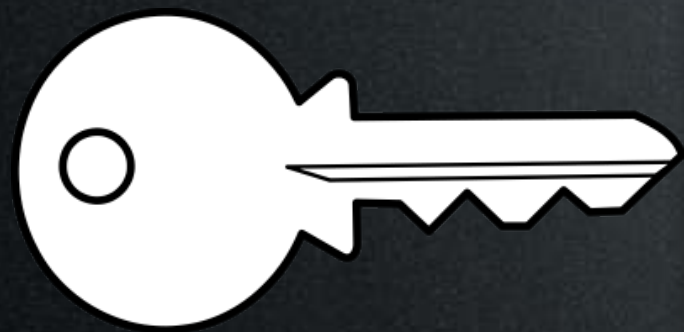
MATT GROENING





Student ID

Test  
Score

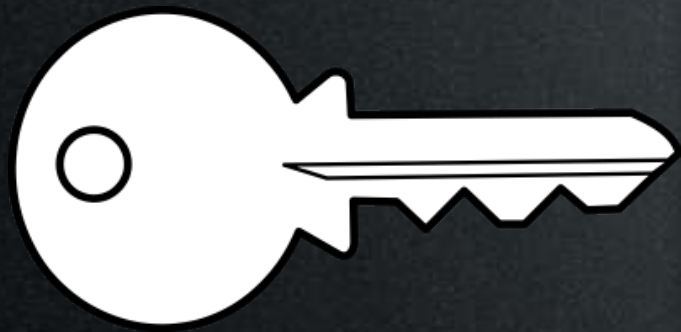


Student ID

Test Score

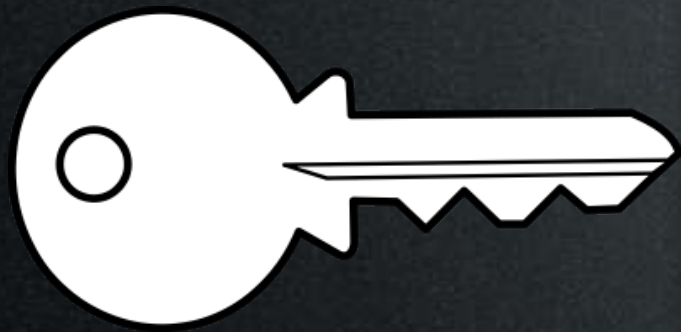
Embed  
Other Stuff





Student ID

Test Score Test Question



Student ID

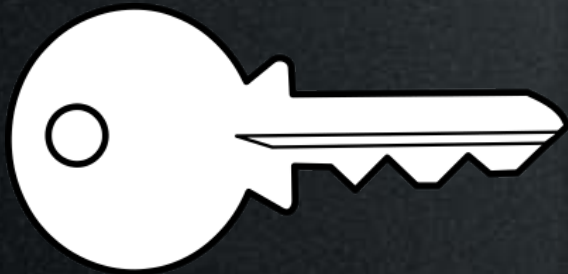
Test Score

Test Question

Concept  
Alignment



# OUT OF SCREEN ERROR!!!

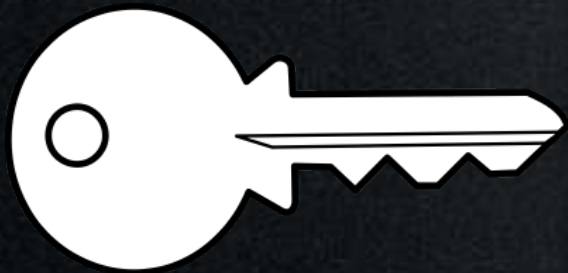


Student ID

Test Score

Test Question

Concept  
Alignment

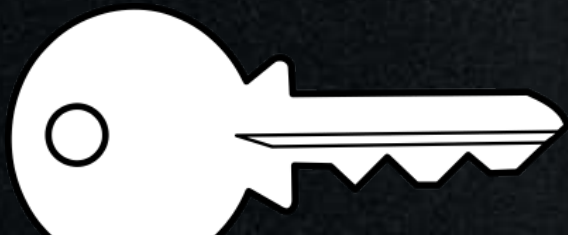


Student ID

Test Score

Test Question

Concept  
Alignment



Test Score

Test Question

Concept  
Alignment



HA HA!



Use a document  
database. It will be...  
Excellent



MAT 63063106

# Assessment 1

## Tiny Tim

Test Question 1

Test Question 2

Test Question 3



Assessment 1  
Tiny Tim

Test Question 1  
Test Question 2  
Test Question 3  
Test Question 4

Test Question 1

$$1 + 1$$

Test Question 2

$$1 * 1$$

Test Question 3

$$1 - 1$$

Test Question 4

$$\frac{1}{1}$$

Assessment 1  
Tiny Tim

Test Question 1  
Test Question 2  
Test Question 3  
Test Question 4

Test Question 1  
**Test Concept 1**

$$1 + 1$$

Test Question 2  
**Test Concept 2**

$$1 * 1$$

Test Question 3  
**Test Concept 2**

$$1 - 1$$

Test Question 4  
**Test Concept 3**

$$\frac{1}{1}$$



Assessment 1  
Tiny Tim

Test Question 1  
Test Question 2  
Test Question 3  
Test Question 4

Test Question 1  
Test Concept 1

1 + 1

Test Question 2  
Test Concept 2

1 \* 1

Test Concept 2  
Multiplication

Test Concept 3  
Division

Assessment 2  
Tiny Tim

Test Question 1  
Test Question 3  
Test Question 4  
Test Question 2

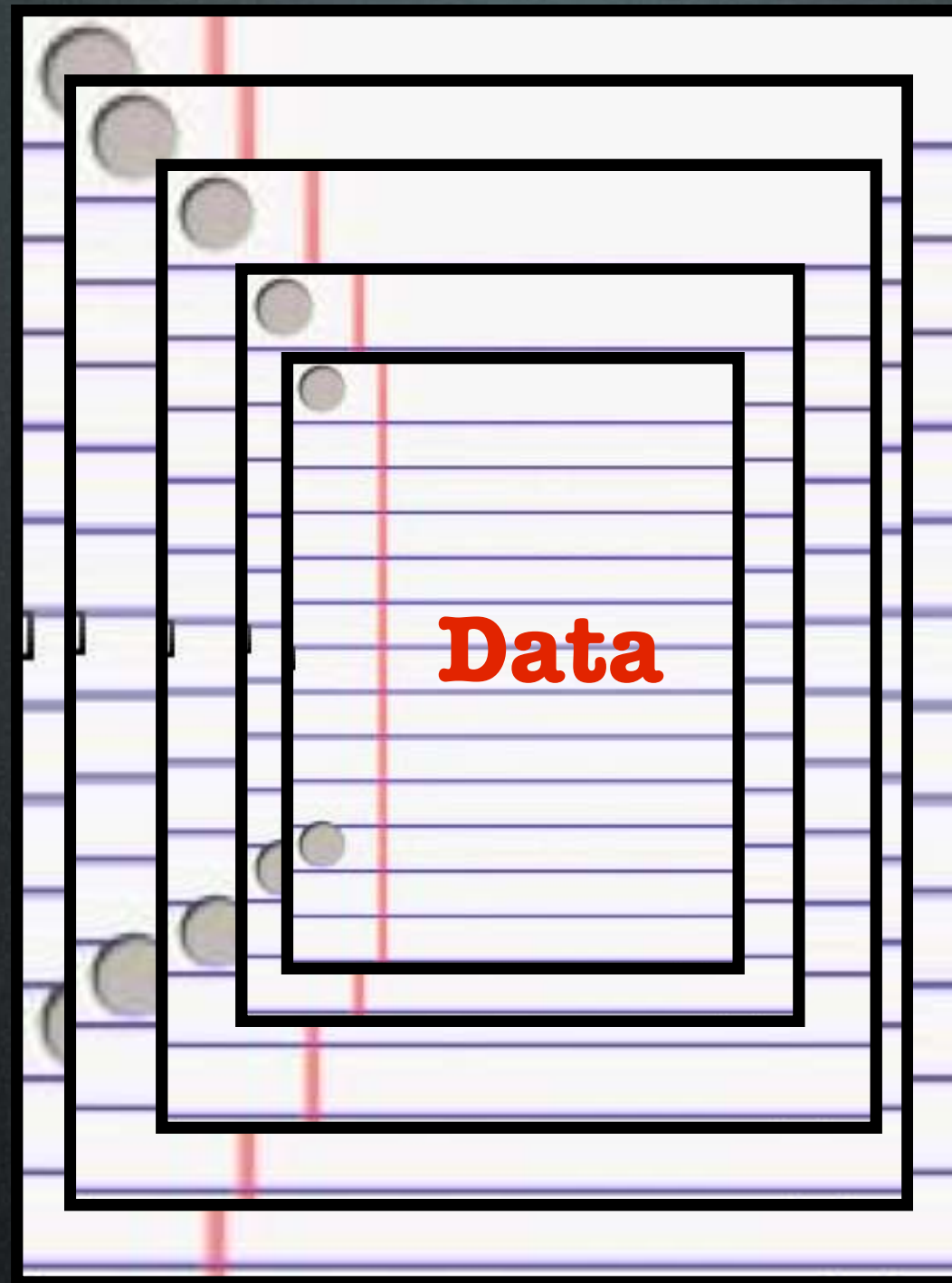
Test Question 3  
Test Concept 2

4 \* 8

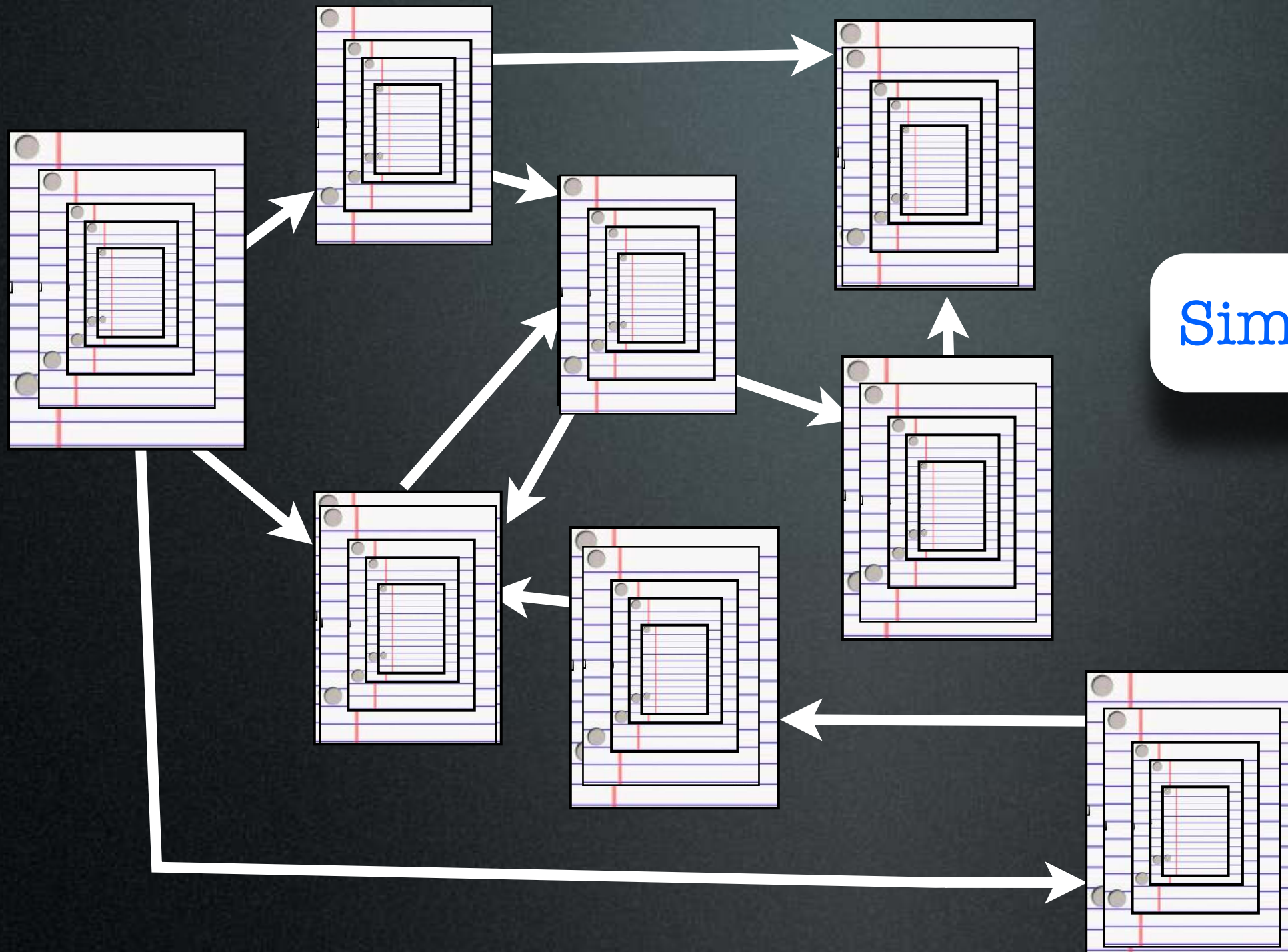
Test Question 4  
Test Concept 3

$\frac{1}{1}$

Test Concept 1  
Addition







Simple!



What the...!



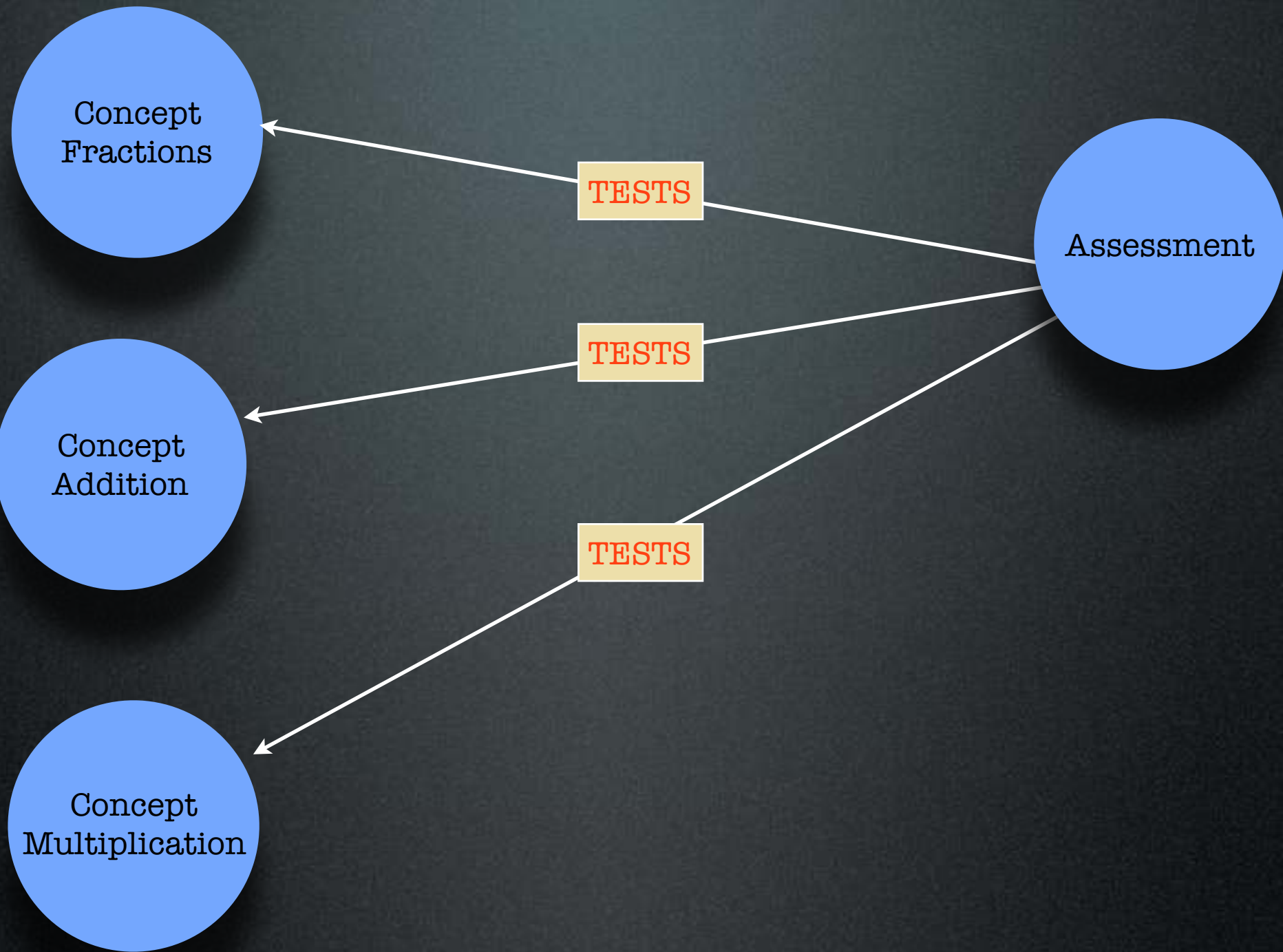




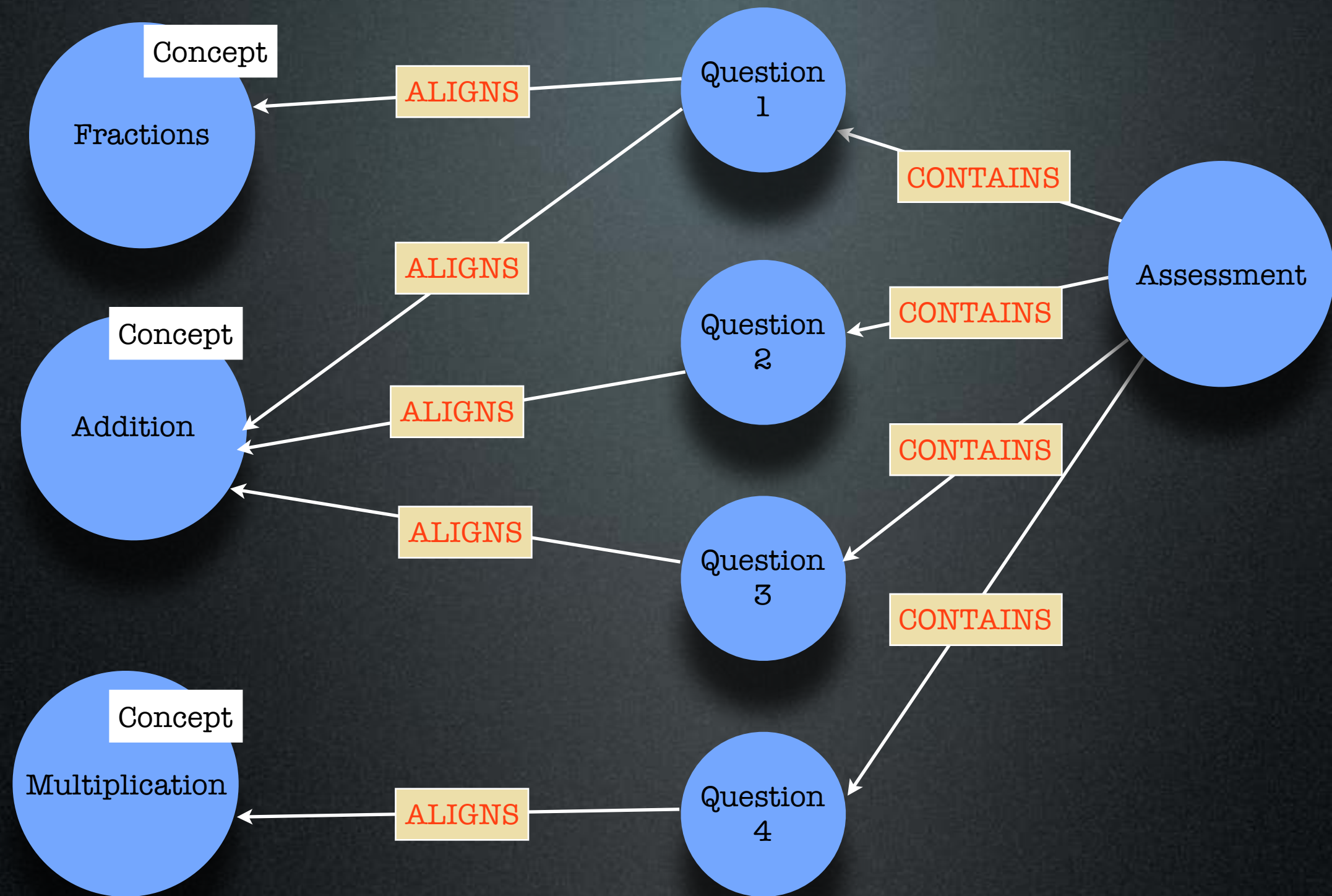
Seems like an index  
free adjacency problem  
[0,1,1,  
1,0,1,  
1,1,0]

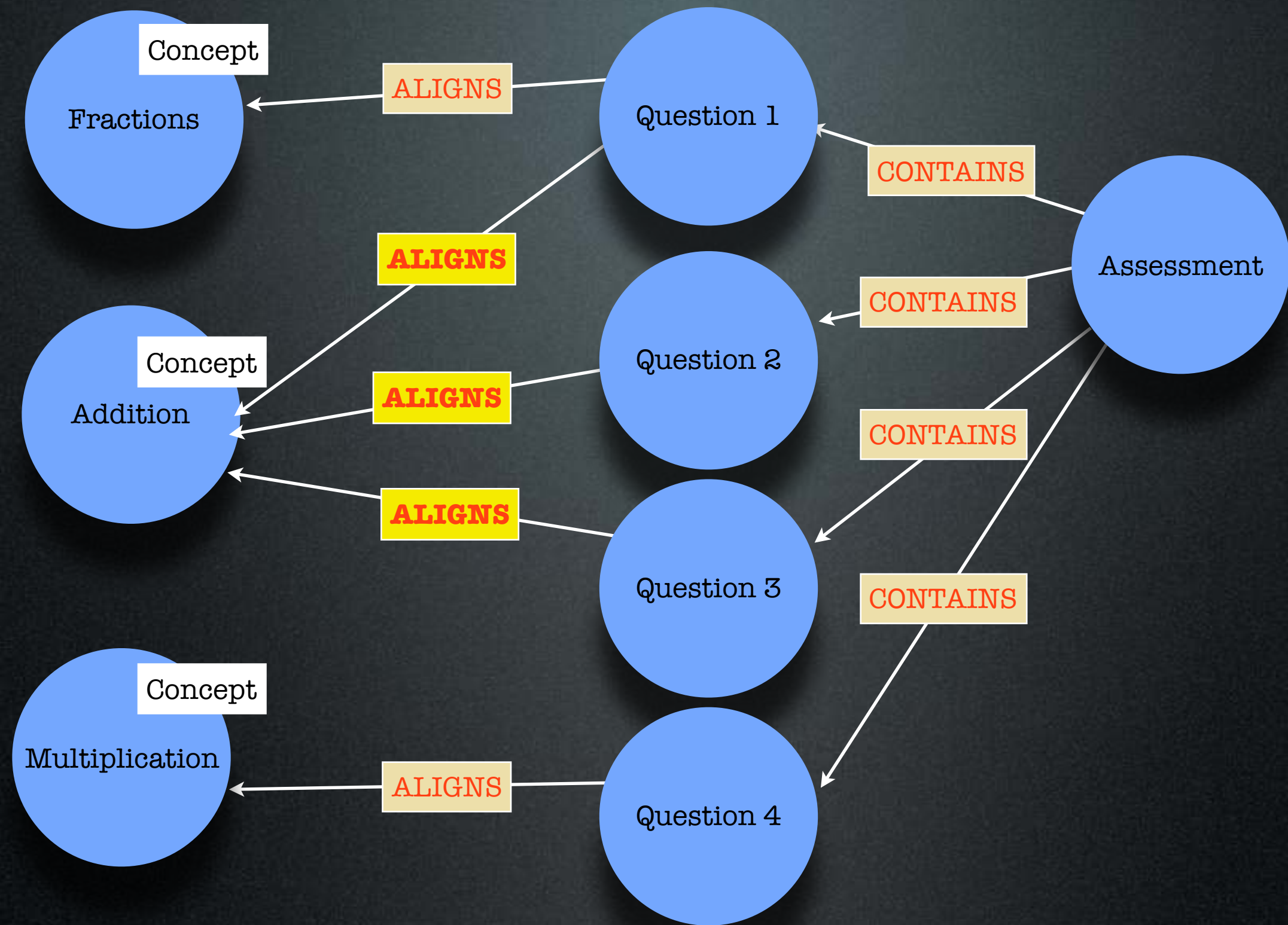


He means a graph  
database, boss

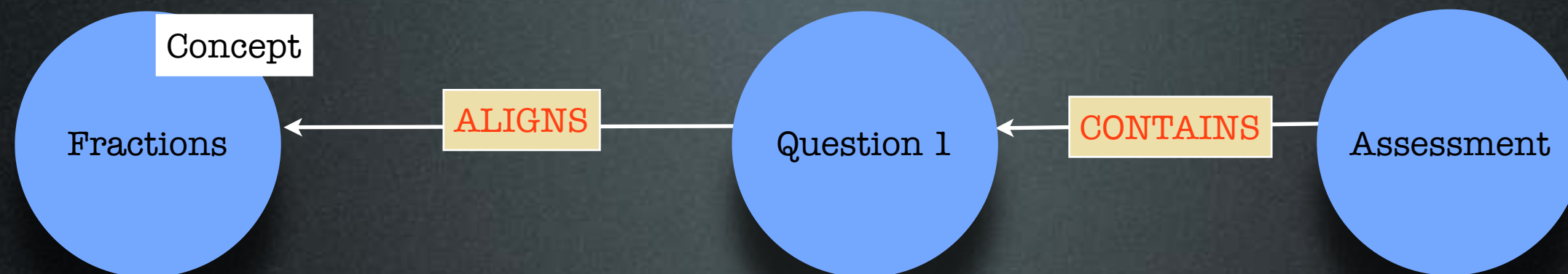


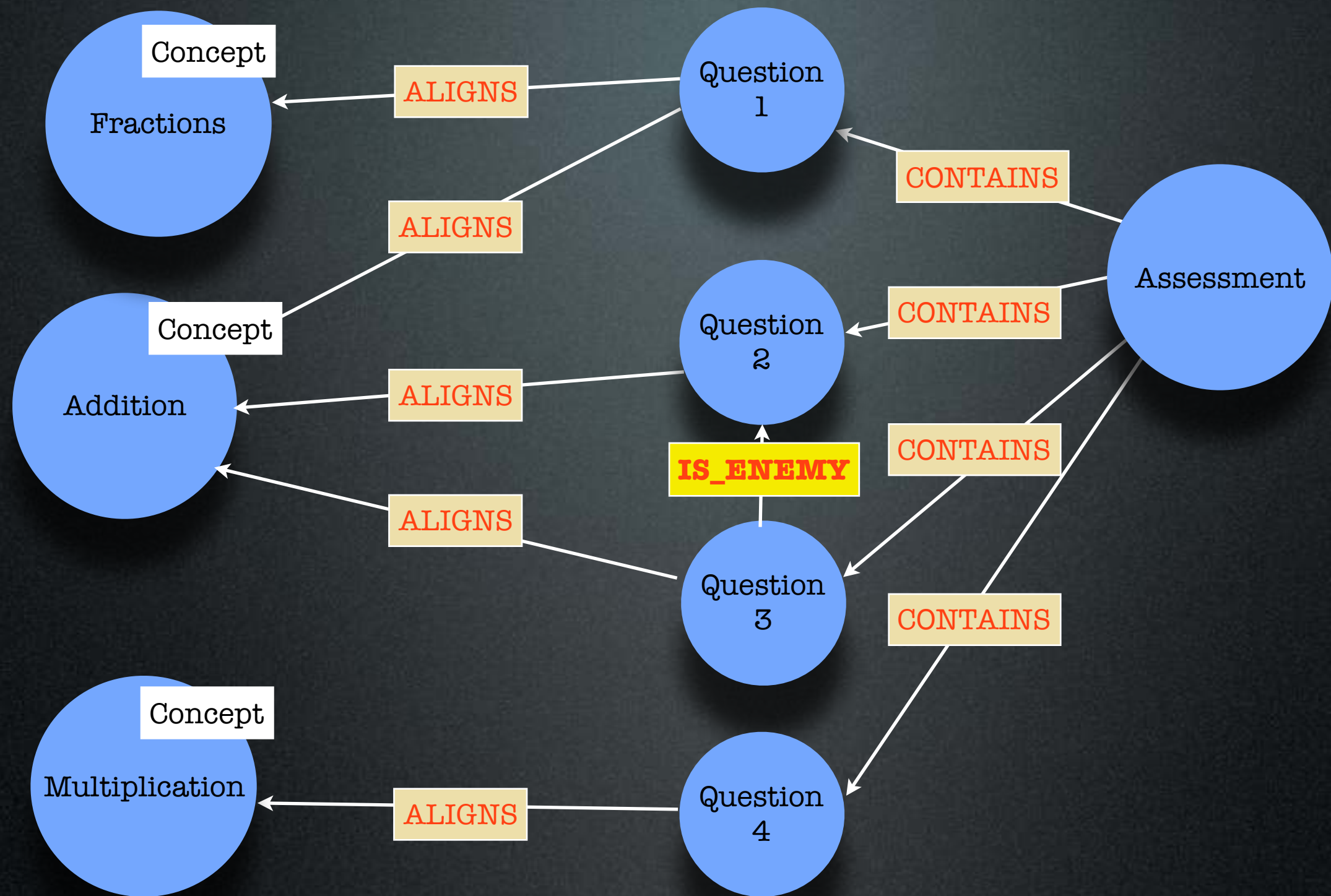














**$120 + 470 =$**

**a) 590**

**b) 420**

**c) 690**





$$120 + 470 =$$

a) 590

b) 420

c) 690



$$120 + 470 =$$







$$120 + 470 =$$

a) 590

b) 420

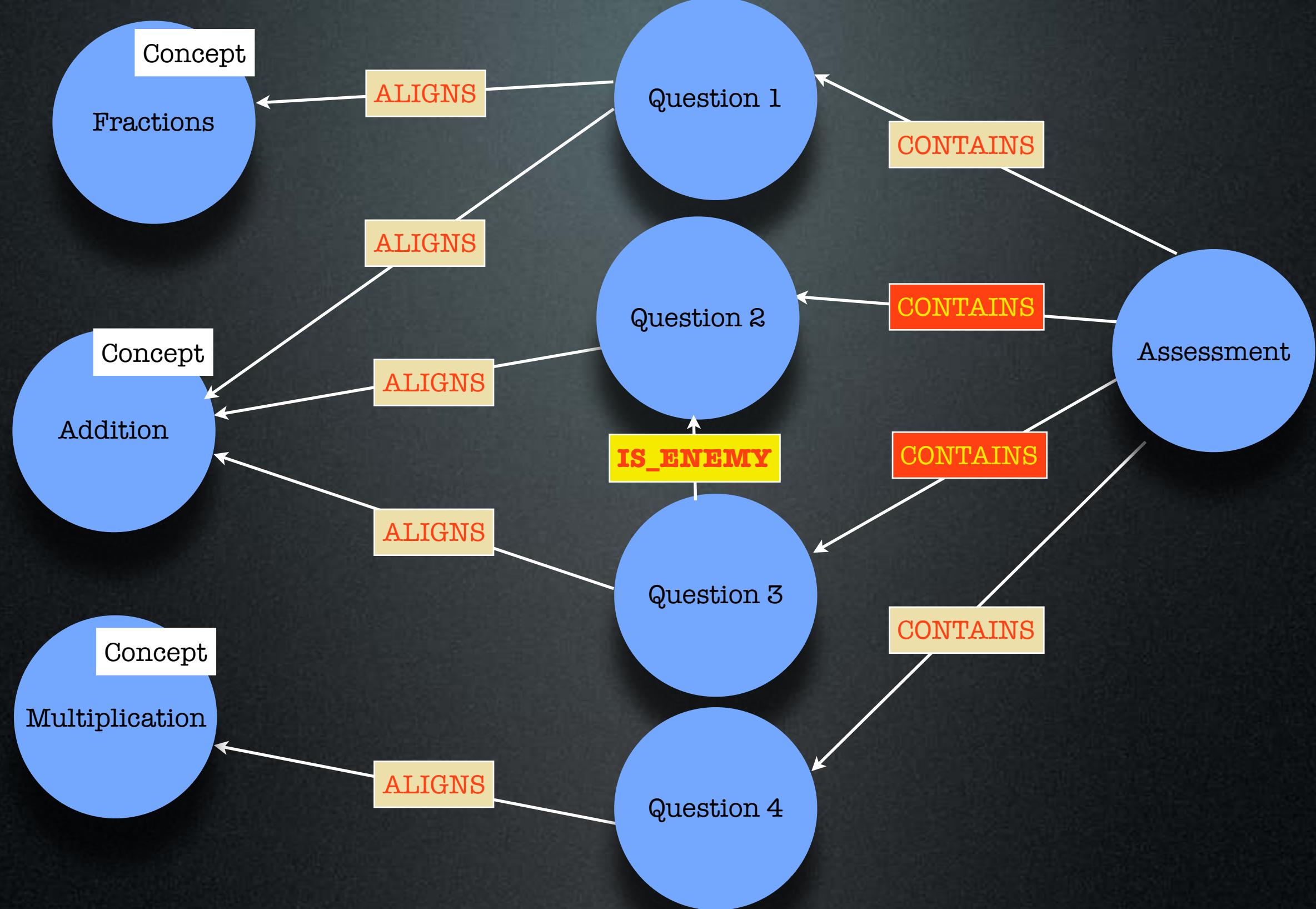
c) 690



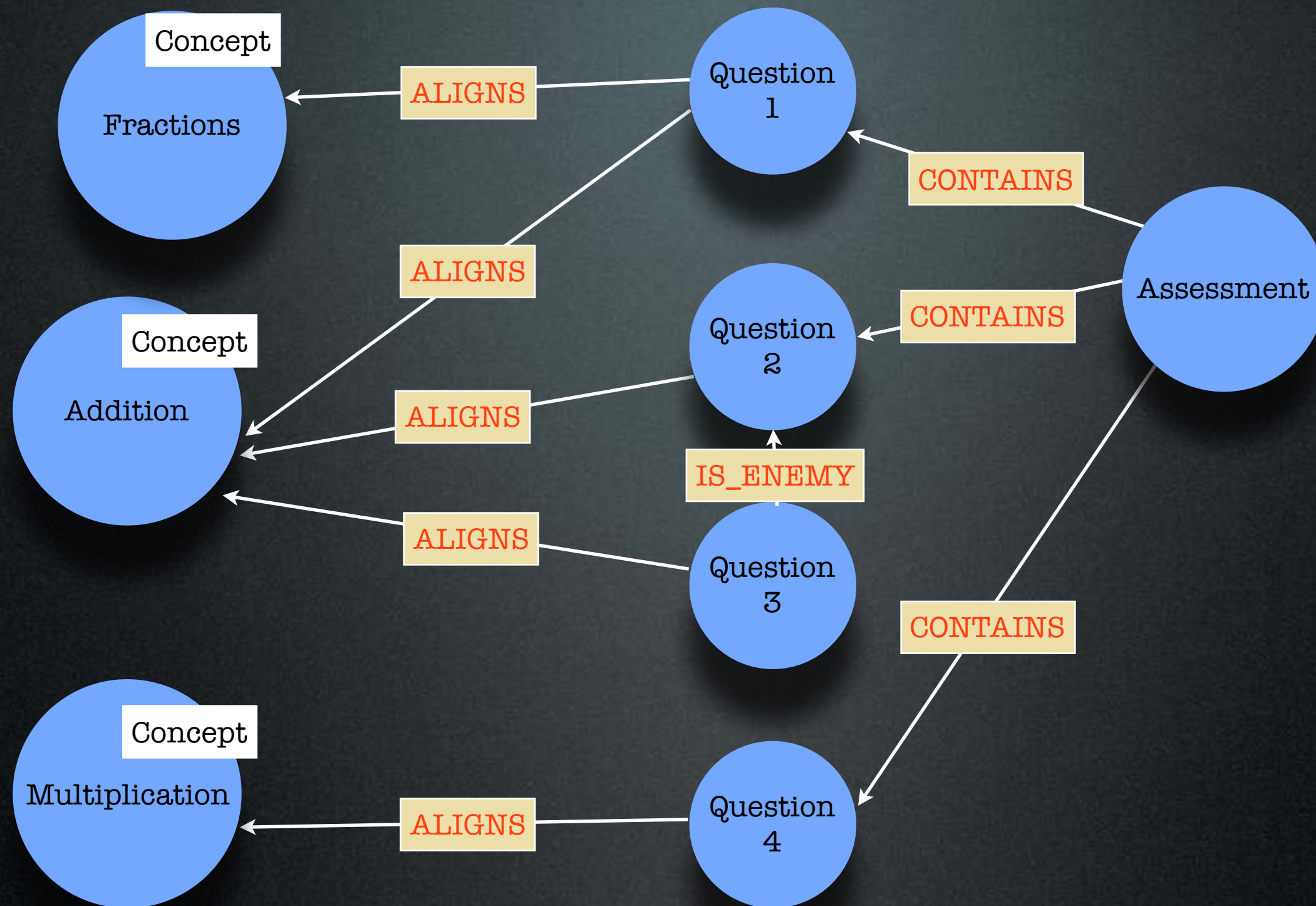
$$120 + 470 =$$

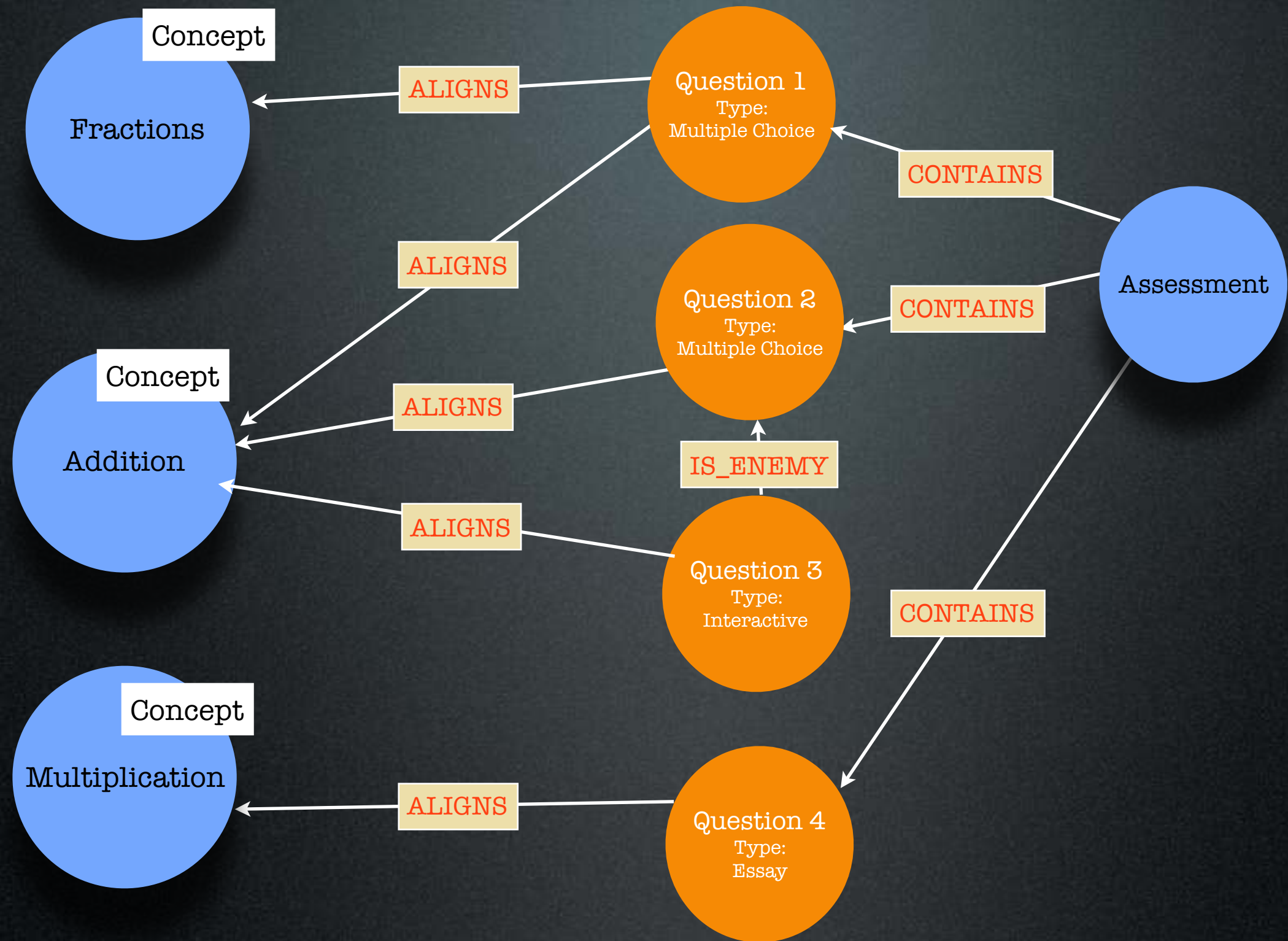
**ENEMIES!!!**



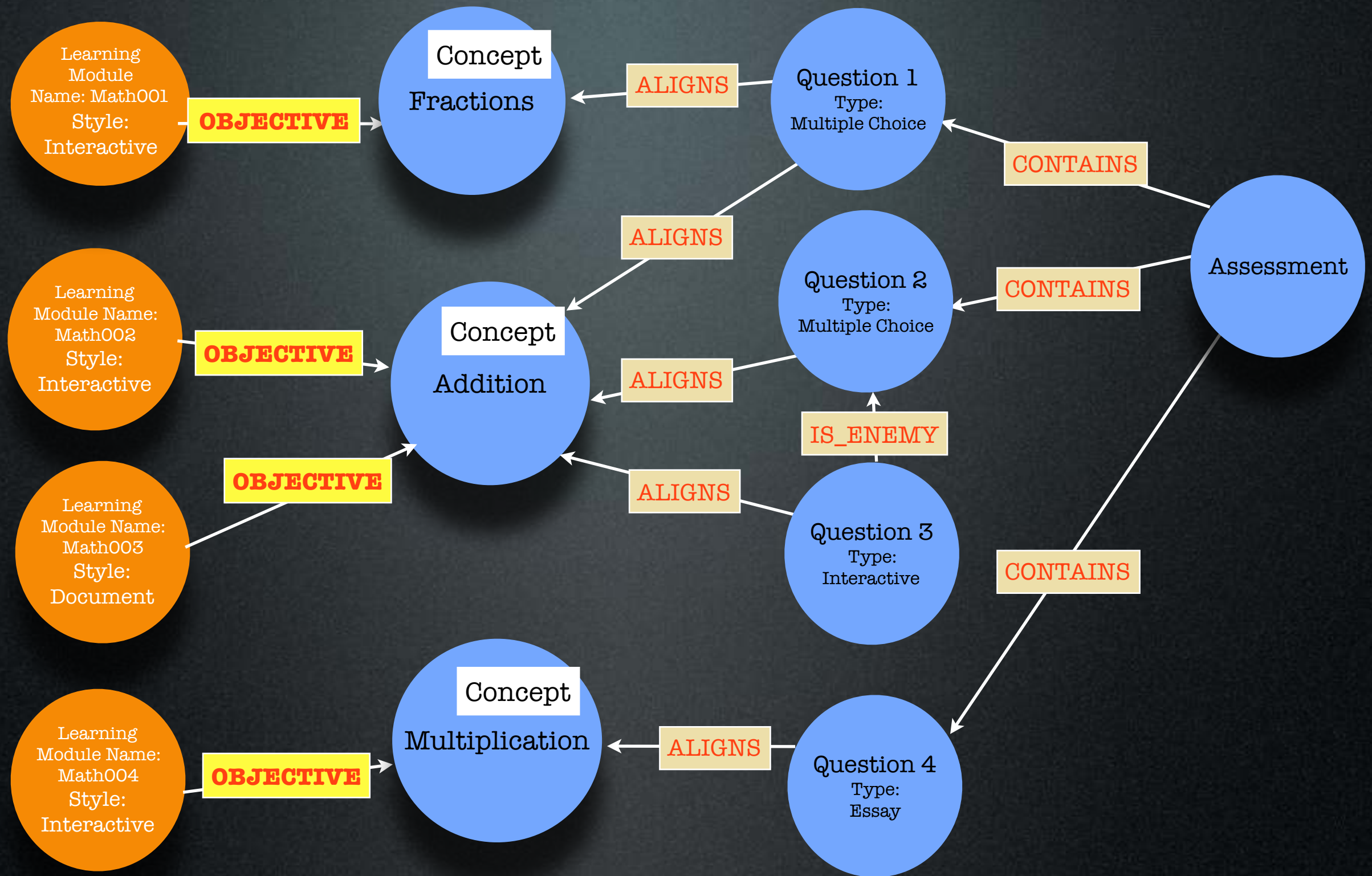


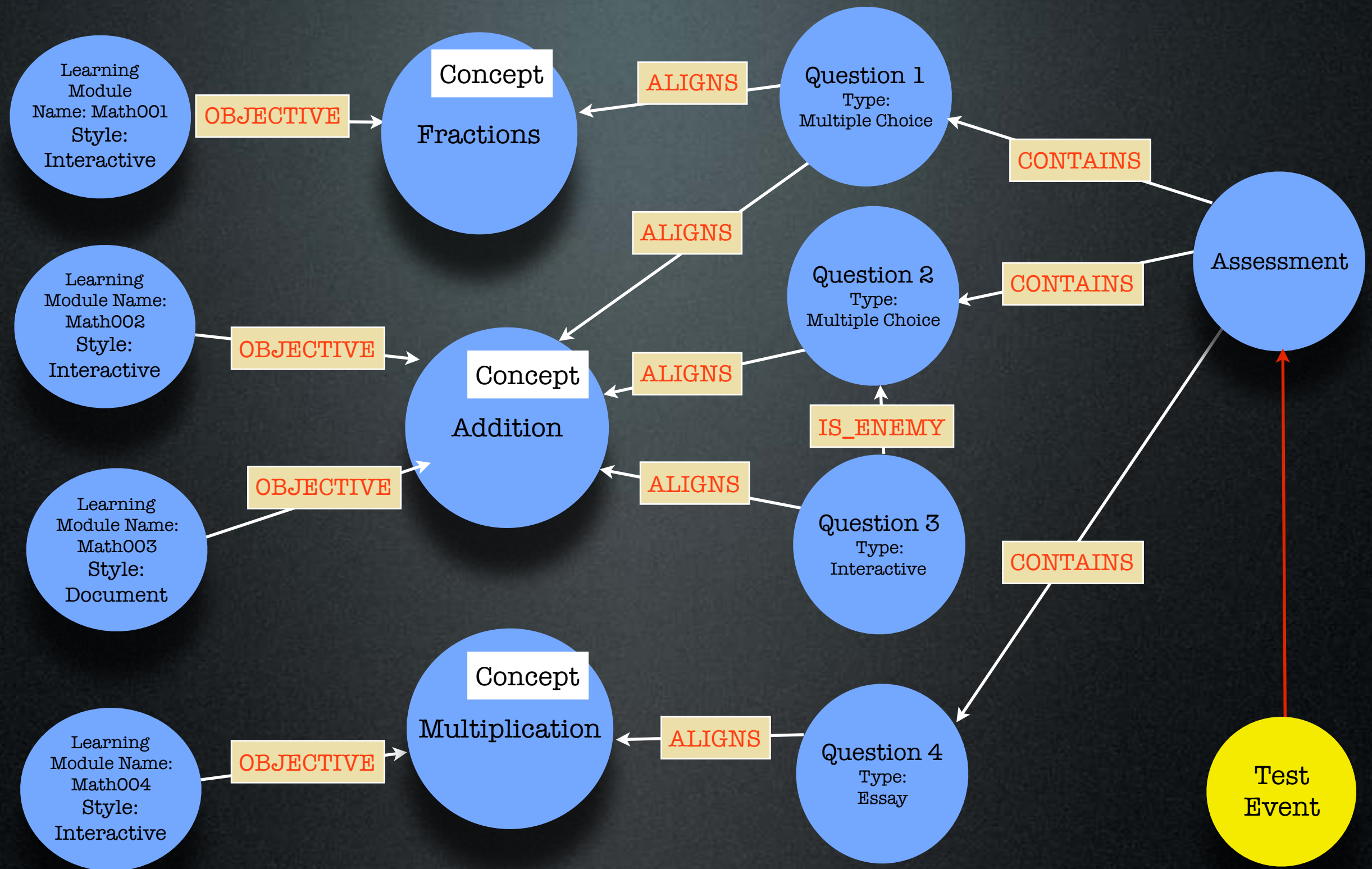




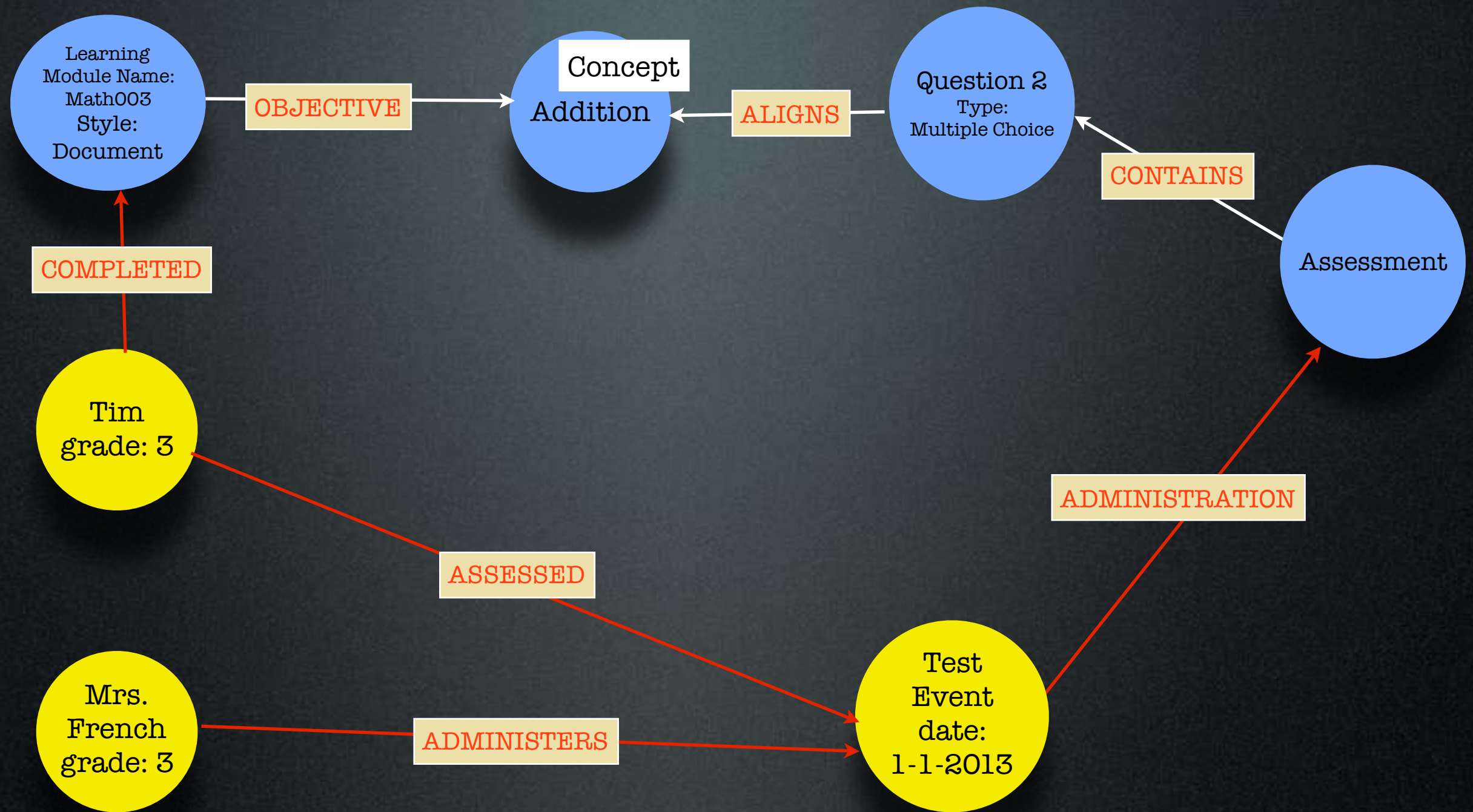


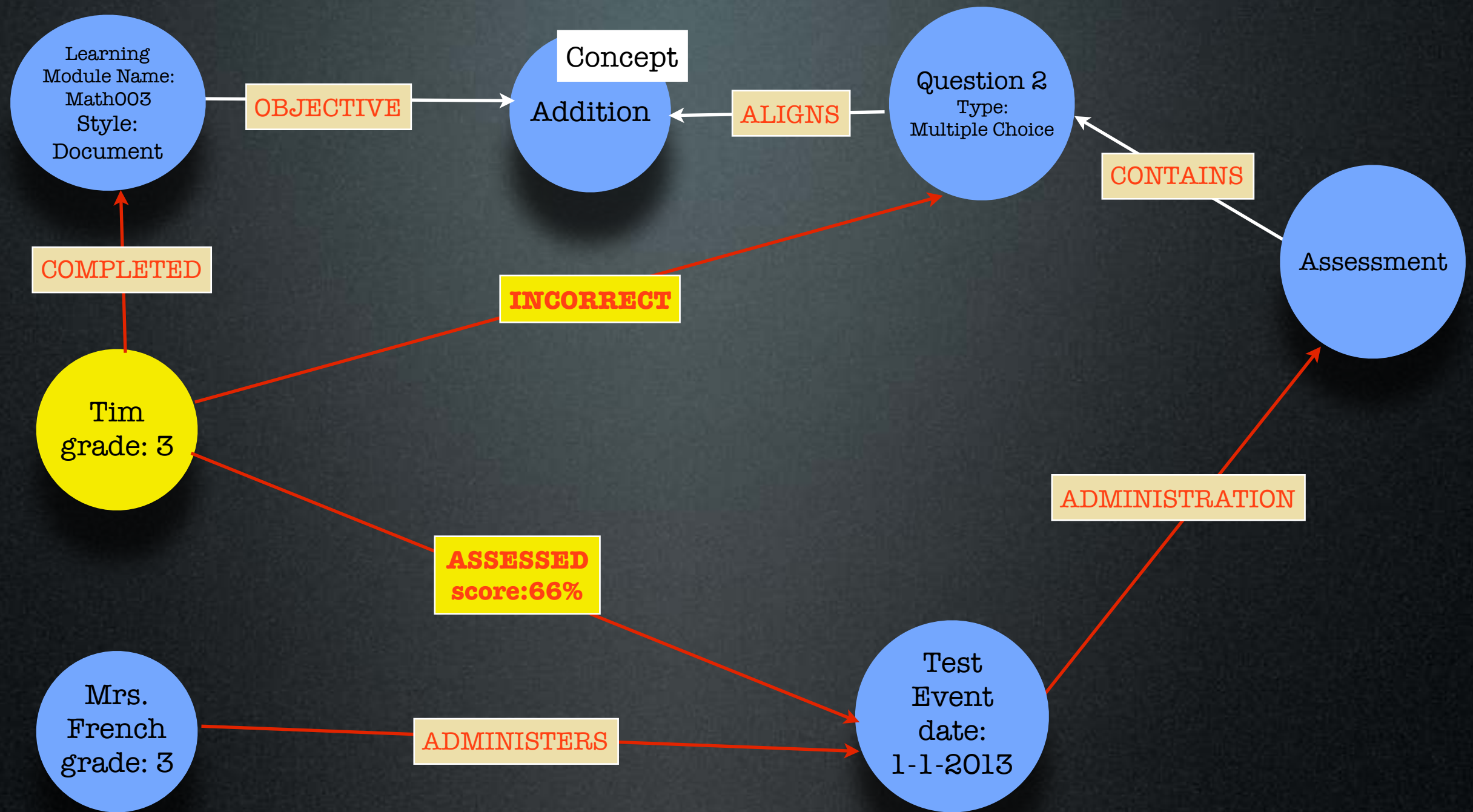




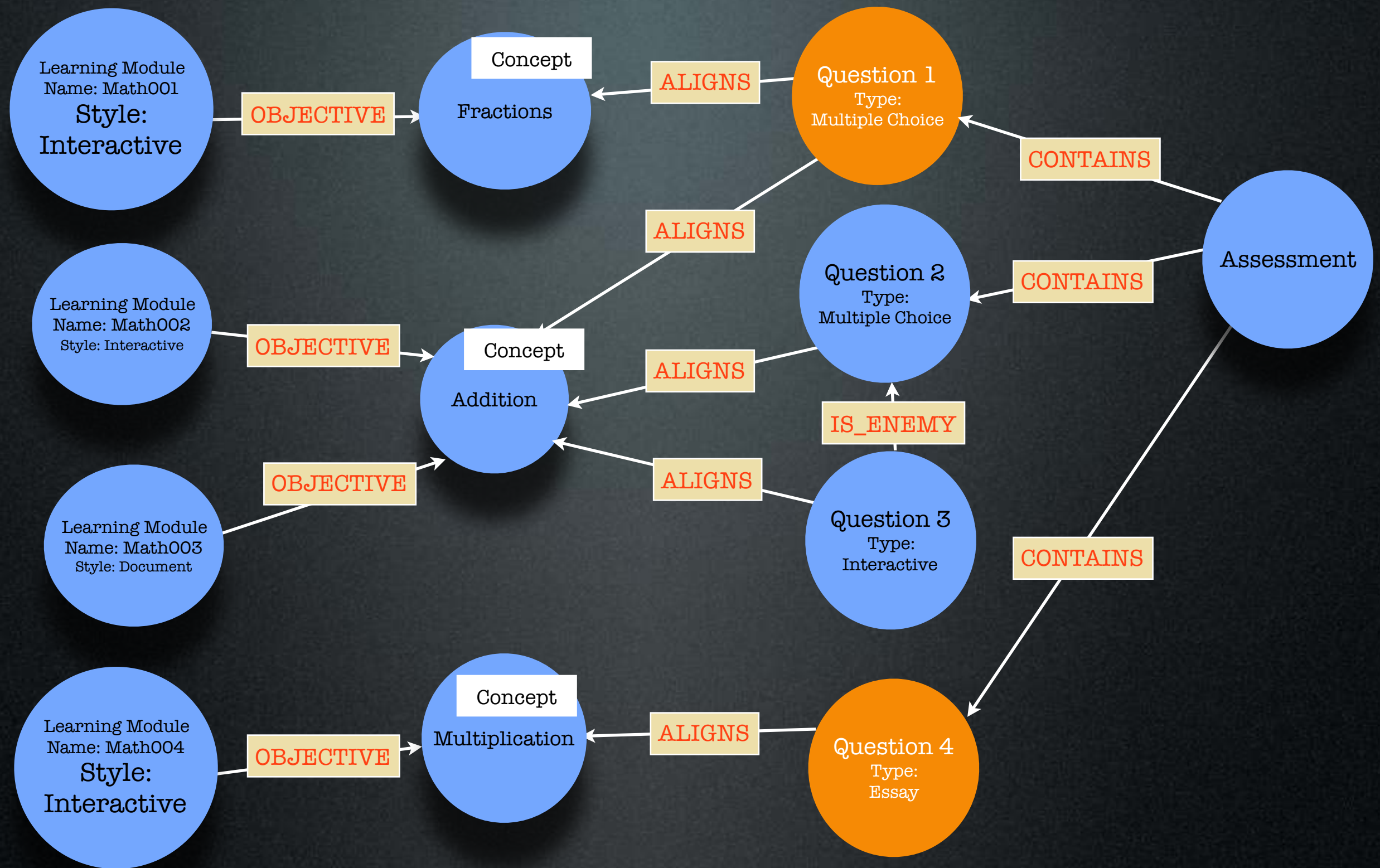


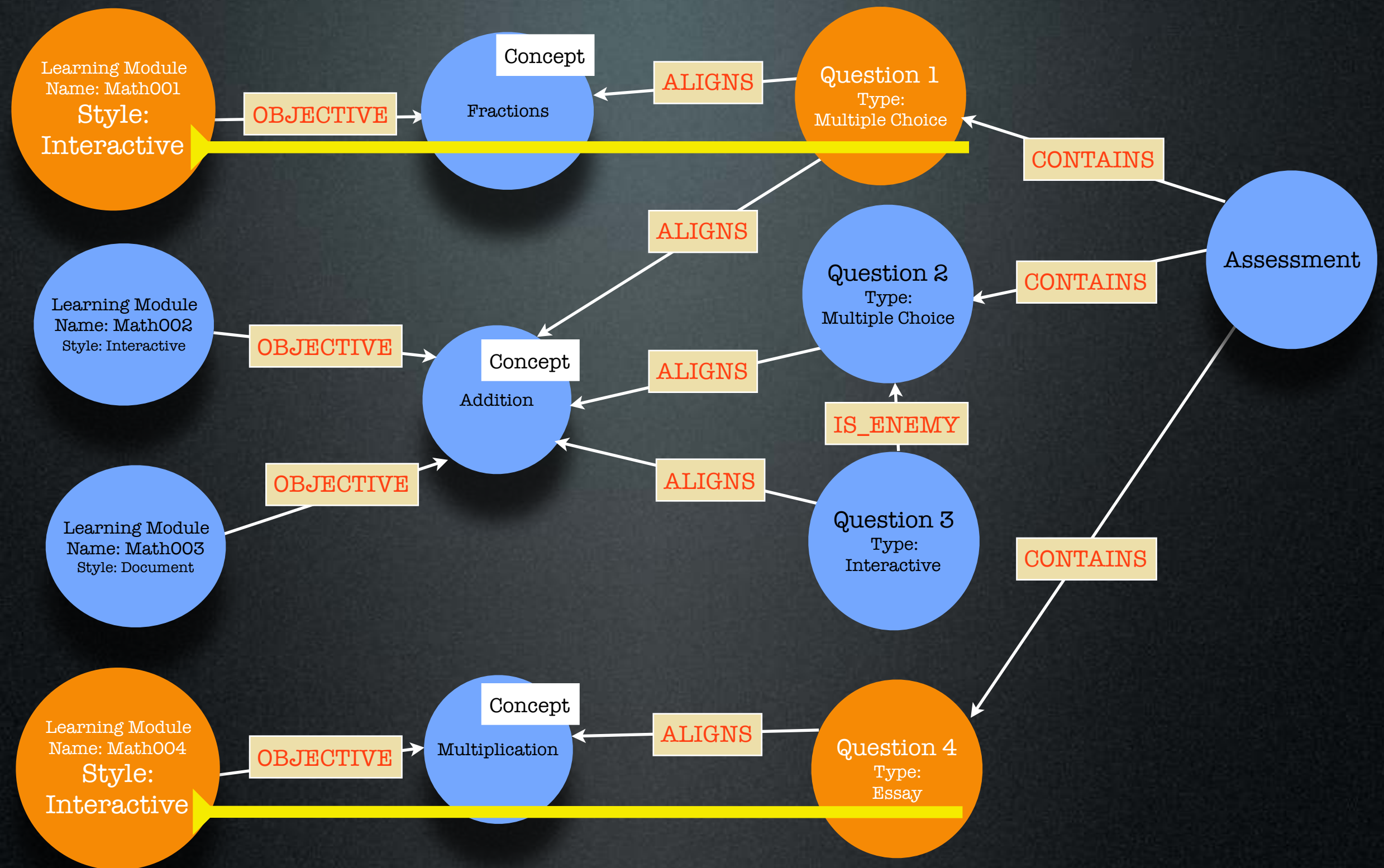




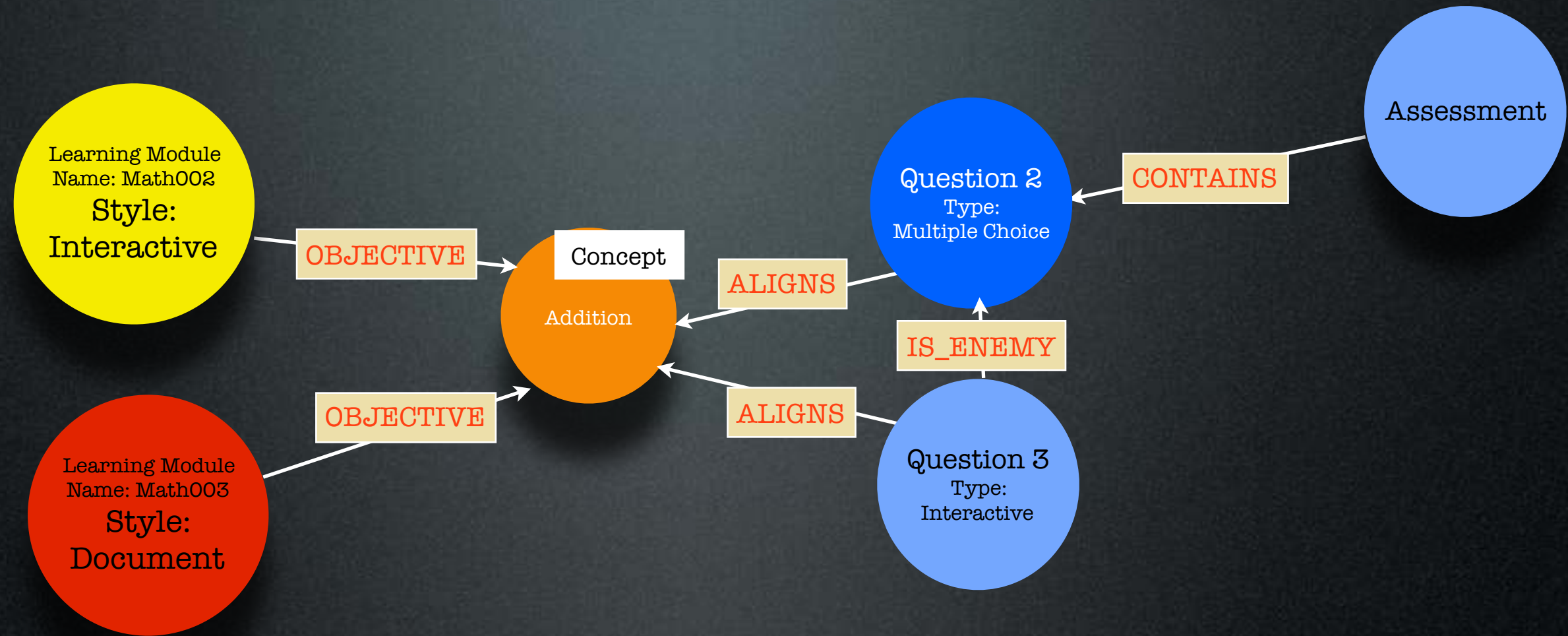


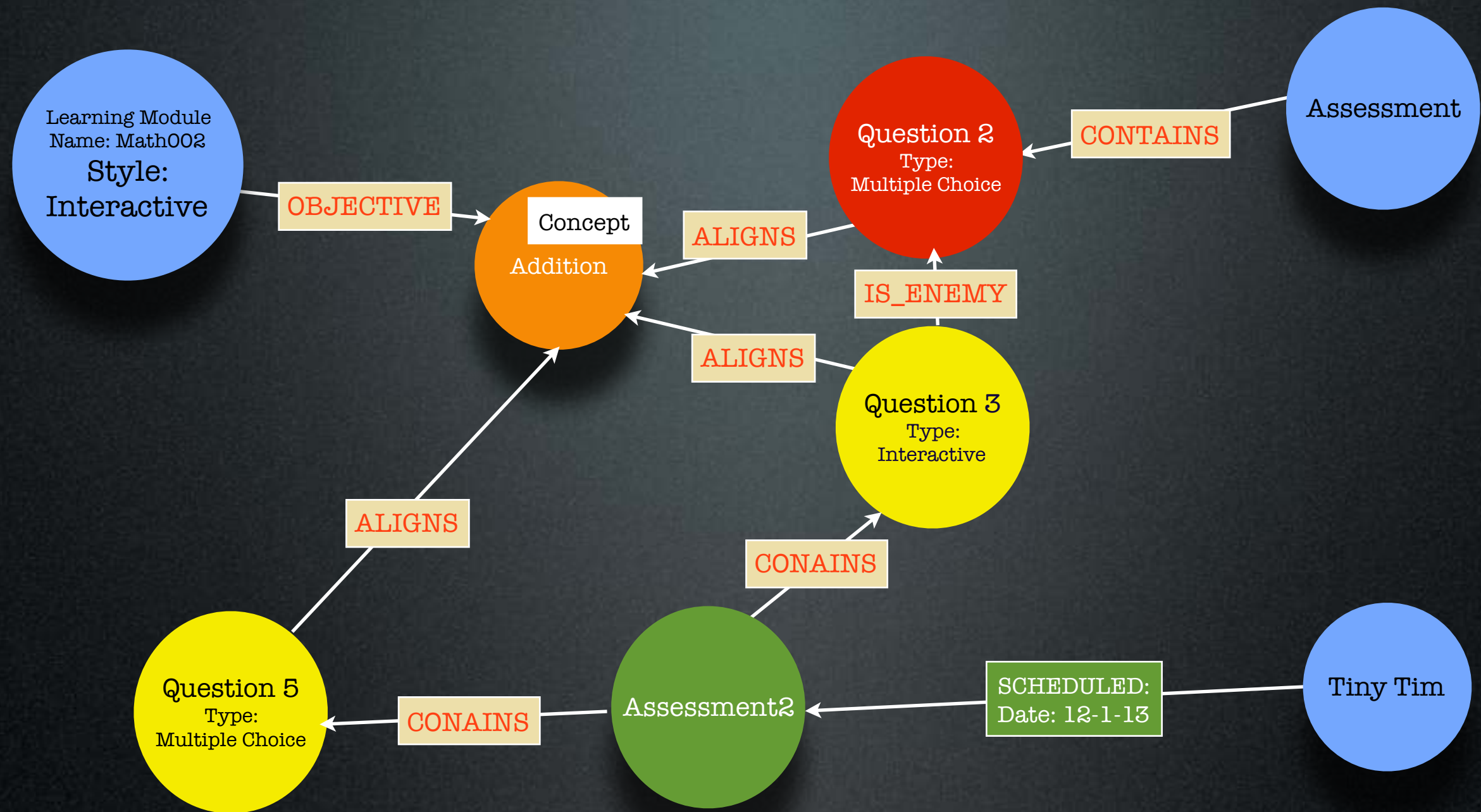














Like I said, boys, a  
graph database! Let's  
see how it works.



# Come for the connected data, stay for the Cypher

<http://localhost:7474/browser/>





Data storage is no  
longer an  
afterthought

LEARNING MODULE RECOMMENDATION...  
CLASSROOM ASSIGNMENT...  
TEST CREATION...  
ADAPTIVE TESTING...  
STUDENT PAIRING REFERRAL...  
OTHERS?



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